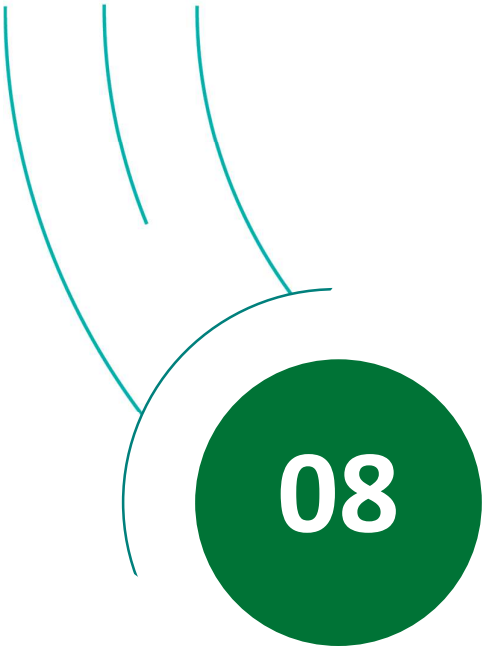
GLENCORE

Maintenance Work Management Guideline

6. WORK MANAGEMENT IMPROVEMENT



Date, Version	27/01/2022, v1.0
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Comment/s	Published

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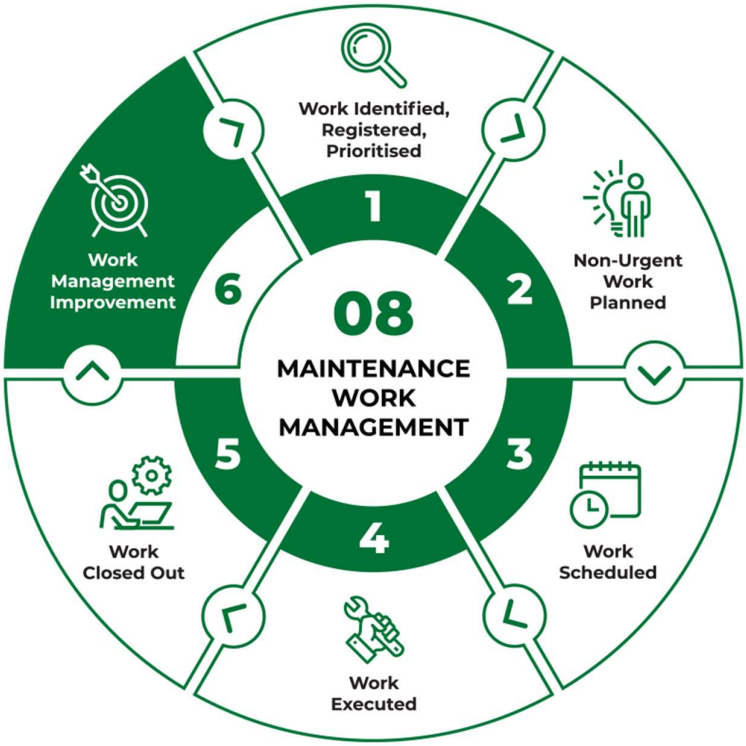
Sub Process Overview

Purpose

The Glencore Copper Asset Management Framework (AMF) enables the business to deliver its Asset Management purpose and vision, as defined in the AMF Policy. The Framework consists of 18 Elements, detailed in Manuals that define how Glencore Copper manages the whole life of its assets to optimise value and ensure assets are capable of meeting the business and stakeholder objectives.

Maintenance Work Management is Element 08 of the AMF and defines effective work management practices and performance requirements to enable work to be performed safely, efficiently, with precision and when required.

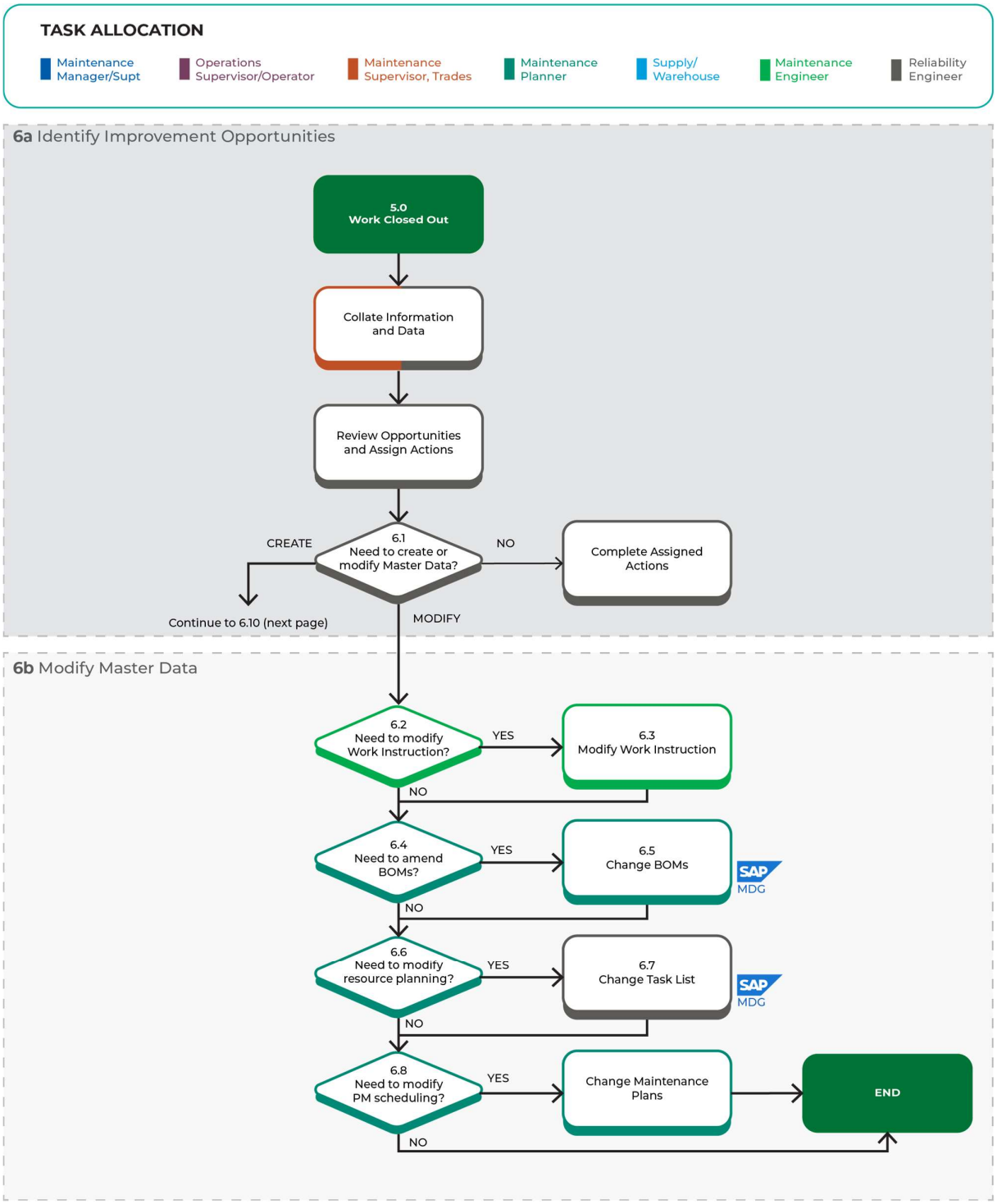
This document is intended to provide end users with understanding and knowledge of sub-process **6. Work Management Improvement**. This includes the analysis of available data and the identification of gaps and opportunities in systems and processes, and the implementation of continuous improvement and standardised best practices to optimise maintenance and achieve increased asset performance.

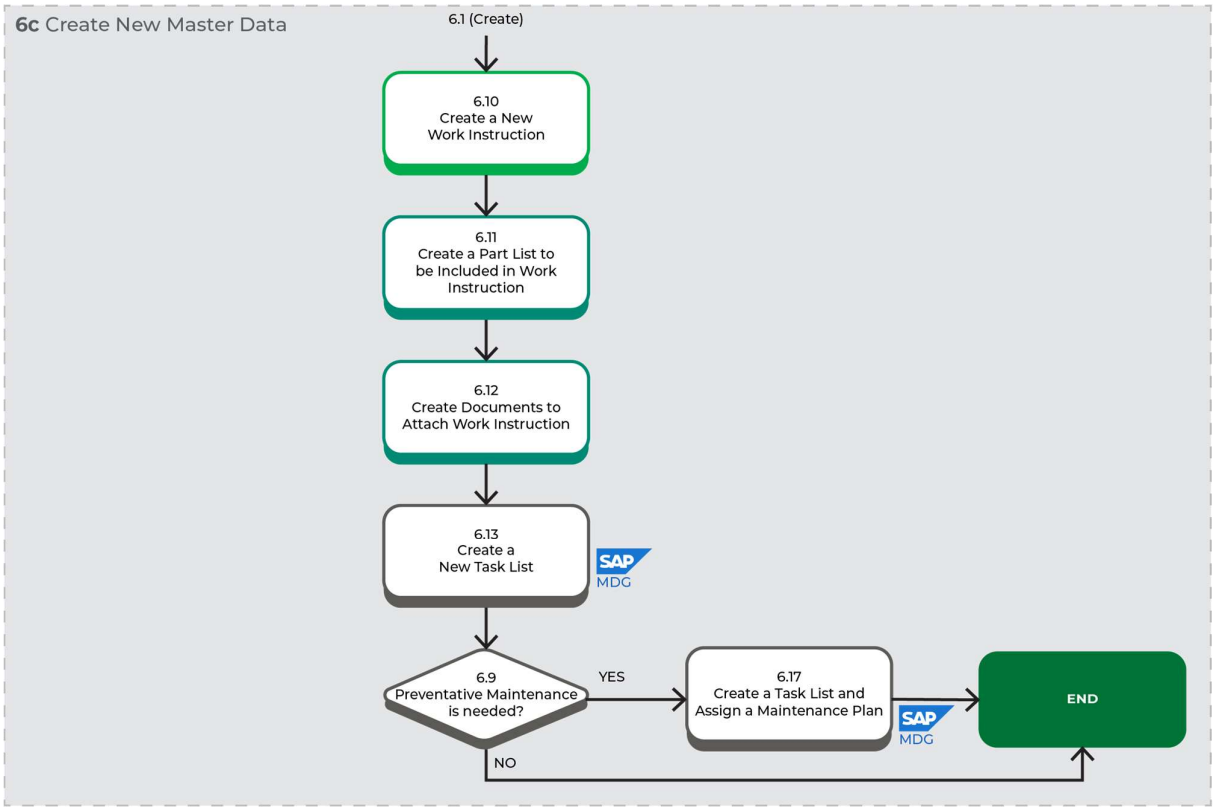


Objectives & Flowchart

The objective of sub-process **6. Work Management Improvement** is to:

- Collate information and data that can indicate gaps in maintenance workflow (system, processes or people).
- Analyse and review this data with the aim of identifying improvement opportunities.
- Implement these improvements in a continuous improvement cycle.


TASK ALLOCATION
Maintenance Manager/Supt
Operations Supervisor/Operator
Maintenance Supervisor, Trades
Maintenance Planner
Supply/ Warehouse
Maintenance Engineer
Reliability Engineer
6a Identify Improvement Opportunities
5.0 Work Closed Out
Collate Information and Data
Review Opportunities and Assign Actions
6.1 Need to create or modify Master Data?
CREATE
NO
Complete Assigned Actions
Continue to 6.10 (next page)
MODIFY
6b Modify Master Data
6.2 Need to modify Work Instruction?
YES
6.3 Modify Work Instruction
NO
6.4 Need to amend BOMs?
YES
6.5 Change BOMs
SAP MDG
NO
6.6 Need to modify resource planning?
YES
6.7 Change Task List
SAP MDG
NO
6.8 Need to modify PM scheduling?
YES
Change Maintenance Plans
END
NO


6c Create New Master Data
6.1 (Create)
6.10
Create a New
Work Instruction
6.11
Create a Part List to
be Included in Work
Instruction
6.12
Create Documents to
Attach Work Instruction
6.13
Create a
New Task List
SAP
MDG
6.9
Preventative Maintenance
is needed?
YES
6.17
Create a Task List and
Assign a Maintenance Plan
SAP
MDG
END
NO

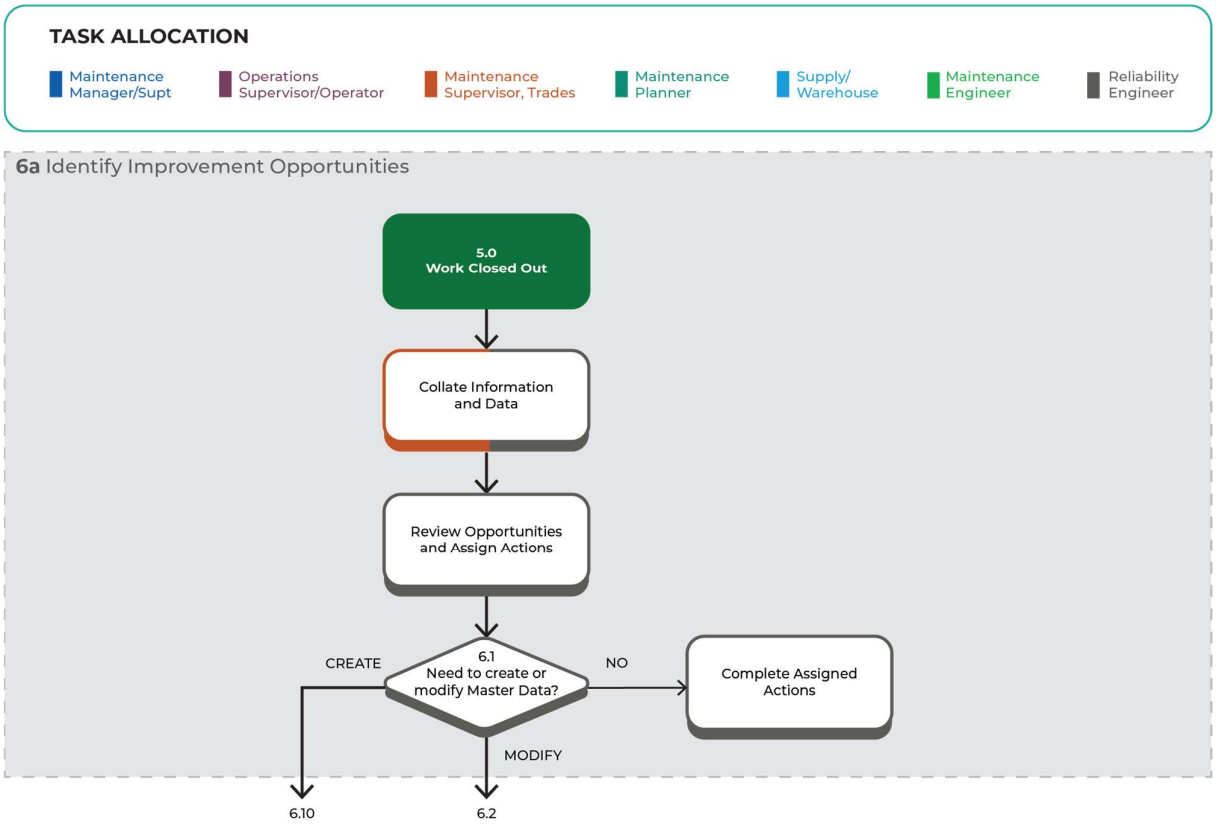
Roles & Responsibilities

R Responsible
S Support
I Inform

		Operations			Maintenance				Other		
		Superintendent	Supervisor	Operator	Superintendent	Supervisor	Trades	Planner	Maintenance Engineer	Reliability Engineer	Supply / Warehouse
	Collate information and data					R		S		R	
	Review opportunities and assign actions		S			S		S	S	R	
6.1	Need to create or modify Master Data?					S		S		R	
	Complete assigned actions		R			R		R	R	R	
6.2 – 6.3	Modify Work Instructions					S	S		R		
6.4 – 6.5	Change BOMs						S	R			
6.6 – 6.7	Change resource planning (Task List)									R	
6.8	Change PM Scheduling (Maintenance Plans)							R			
6.10	Create a new Work Instruction								R		
6.11	Create Parts List to include in the Work Instruction						S	R			
6.12	Create documents to attach to the Work Instruction							R			
6.13	Create a new Task List									R	
6.9	Preventative Maintenance needed?									R	
6.17	Assign a maintenance plan									R	

6. Work Management Improvement

6a Identify Improvement Opportunities



What?	Determine if there are improvements that can be made to Master Data or workflow processes.
Why?	*Master Data is the core data that underpins much of SAP. Having the correct Master Data is absolutely key to effective and efficient Work Management. Master Data is not often changed, and is subject to strict review and approvals workflow to manage and control any changes.* The intent of this step is to identify where Master Data or Work Management processes can be improved or updated. This is done by collating and analysing a range of information and data from the previous week (W-1), with a view to identifying gaps and improvement opportunities, and logging these opportunities into an action log.
Who?	• Reliability Engineer (process lead) • Maintenance Supervisor • Maintenance Planner • Maintenance Superintendent • Maintenance Engineer • Operations Supervisor & Superintendent (optional)

How?

These first steps cover the collation of relevant information and data to be prepared in advance of the **Weekly Improvement Meeting**.

Maintenance Supervisor collates Work Management improvement opportunities (from their review of the Work Order history, refer to sub step **5c Close Work Orders** in sub-process **5. Work Closed Out**). This could include:

- Amended Bill of Materials (BOM) and/or parts lists.
- Updates to Work Instructions and/or Task Lists (added job steps, improved details or instructions etc).
- Process failings and inefficiencies (e.g. kitting of parts not performed, job packs missing, equipment isolation delays, equipment not cleaned, permit delays etc.)
- Amended job duration estimates.
- Possible changes / requests for investigation into maintenance strategies.

Reliability Engineer to generate and analyse reliability data and trends such as Downtime analysis, MTBF, MTTR etc. Refer to **E05 Equipment Operating and Maintenance Strategy** for further details.

Maintenance Planner to generate a set of relevant Maintenance Key Performance Indicator (KPI's). Refer to **SAP PM – Reporting Information** and SAP PM QRGs for the individual KPIs and Reports for further guidance.

*Note: These following steps to review opportunities are completed in the **Weekly Improvement Meeting** run by the Reliability Engineer. Refer to meeting outline at the end of this document for further guidance.*

Reliability Engineer to lead the discussion and analysis of performance, using the following data:

- Opportunities identified by Tradespersons during the execution of work (refer above).
- Maintenance KPI data.
- Reliability Data (downtime analysis, MTBF, MTTR, repeat failures etc).
- Other relevant Root Cause or Defect Elimination studies.

During the analysis, the following must be considered:

- Does existing Master Data need to be modified (e.g., Equipment Structure (FLOC), BOM's, inspection frequency, Work Instructions, Task Lists, task durations etc)?
- Is a new maintenance strategy needed (e.g. a new service or inspection routine)?
- Are any modifications to equipment or assemblies required?
- Are there any issues with Work Management workflow, such as the planning and scheduling sub-processes (e.g. blurred responsibilities, non-attendance at meetings), not executing to schedules etc?
- Review action plan from the previous meeting.

Develop and update the action plan confirming outcomes from the above review. The action plan must identify what the action is, who is responsible for completing it, and when it is due.

- Many of the above actions (e.g. changes to Master Data, new or revised maintenance Strategies, equipment modifications etc), must also be considered with regard to **E05 Equipment Operating and Maintenance Strategy**.
- Where equipment modification has been recommended, refer to the **E05 Equipment Modification Process** (change management).
- Where changes to internal processes have been recommended, refer to Maintenance Superintendent for guidance.

The following example for a sample of a simple spreadsheet Action Plan template. The action plan must clearly indicate the action description, who is responsible, the due date, the actual completion date, and progress notes.

EHM Mechanical Maintenance System Improvement Meeting

ACTION PLAN

Meeting held weekly Wednesday 1pm in Main Conference Room

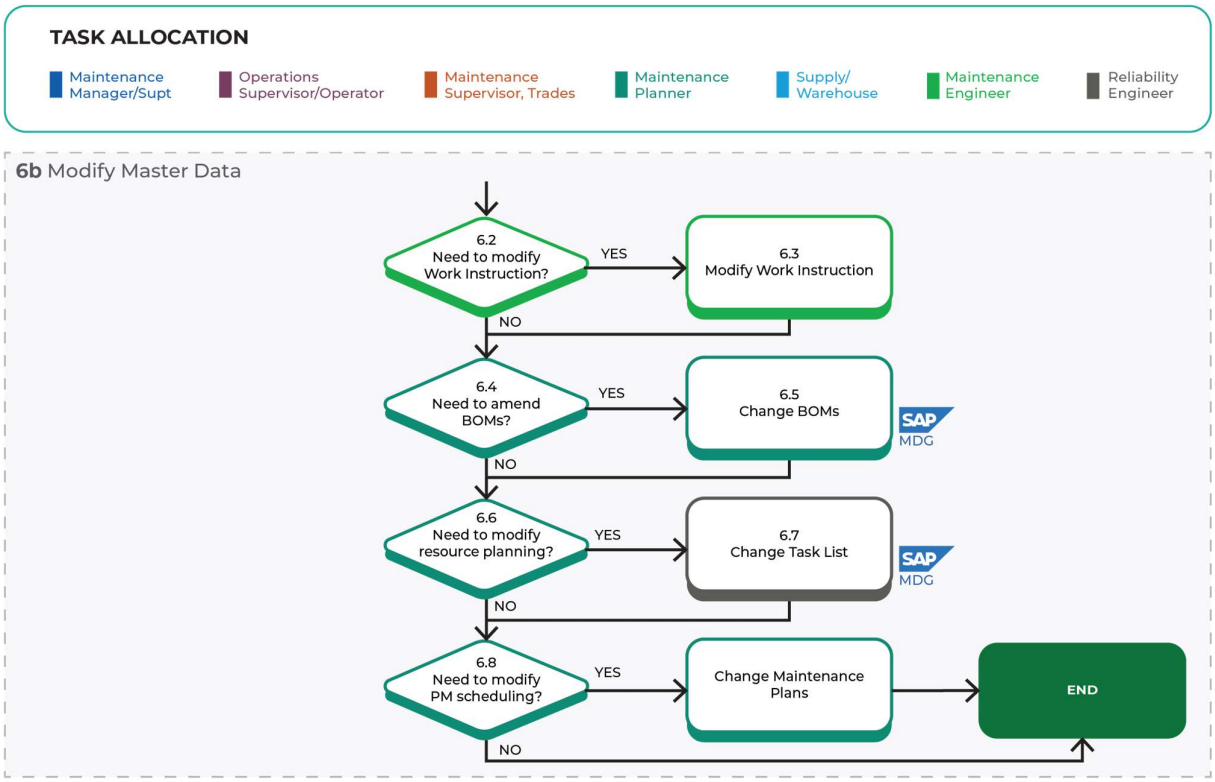
Action Plan Owner: B Smith Reliability Engineer

Action #	Date Raised	Description	Responsible	Due Date	Completed Date	Progress Comments
1	21/07/2021	Update BOM for Pump#5 , refer WO 206542 for details	Jane Doe	3/08/2021	3/08/2021	
2	28/07/2021	Revise Task List for Sag Mill monthly inspection to include lube system checks	Joe Smith	1/09/2021		Waiting feedback from Lube supplier
3	28/07/2021	Raise Change Management request for trunnion plinth modifications	Bill Zane	15/08/2021	14/08/2021	
4	1/08/2021	New 3 monthly mechanical inspection of conveyor 8 system	Joe Smith	2/08/2021		Still developing scope. Expect completion by end August
5	1/08/2021	Enforce operations people attendance at Weekly Planning Meeting	Simon Boss	1/09/2021		send out email, waiting to assess effectiveness.
6	8/08/2021	Change frequency of Kiln walkway inspection to 12 monthly	Joe Smith	20/08/2021		MDG request raised, still waiting on review and approval. Will chase up reviewer 21/8/21

When?

Weekly, before and at the **Weekly Improvement Meeting**.

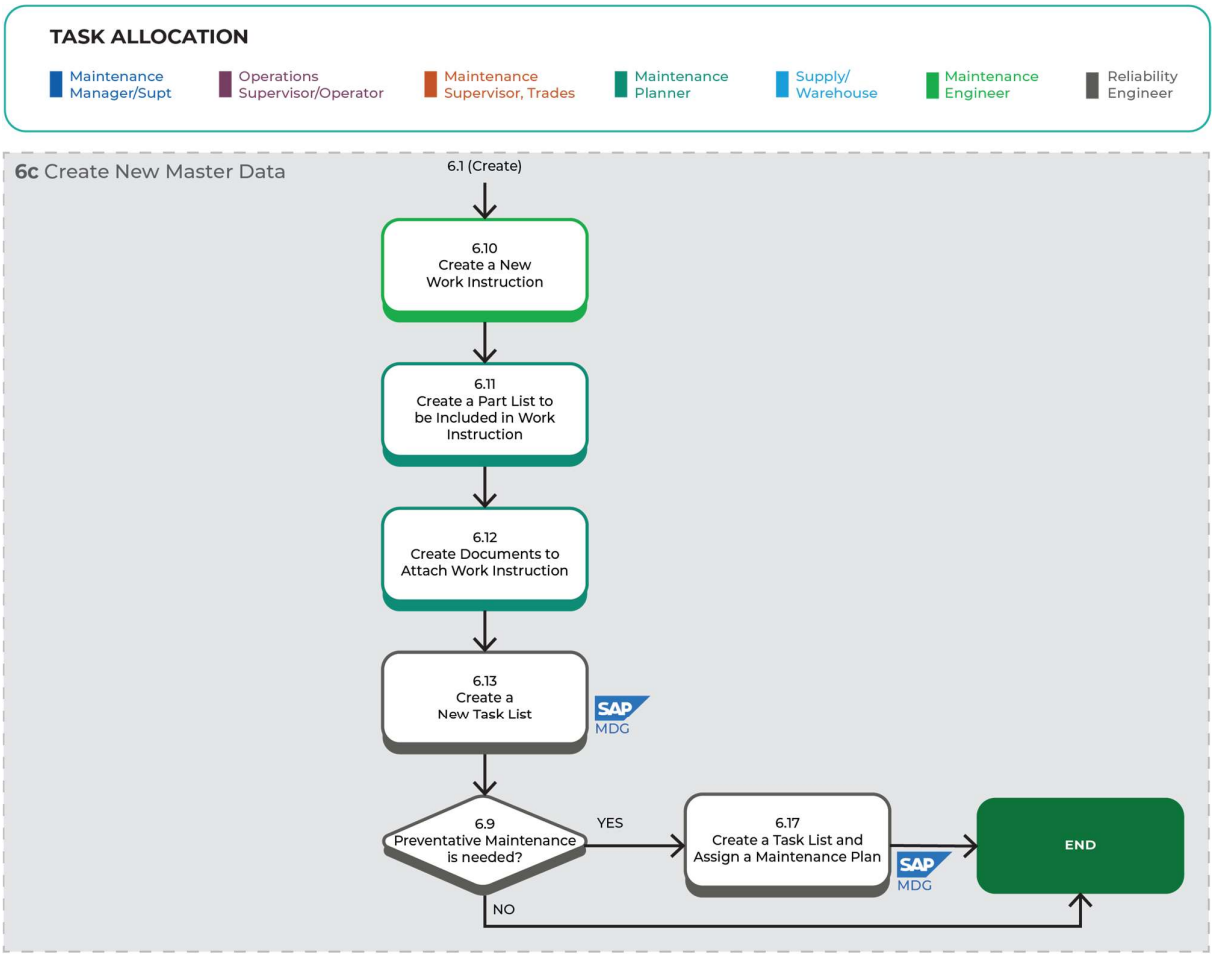
6b Modify Master Data



What?	Modify Master Data.
Why?	**Step 6a** identified where Master Data needed to be modified or improved. This step provides an overview of how to modify the Master Data.
Who?	• Reliability Engineer • Maintenance Engineer • Maintenance Planner • Maintenance Supervisor *Note: Responsibility for modifying different types of Master Data lie with different roles. Please refer to the Roles and Responsibilities page in this document for further guidance.*
How?	*Note: All Master Data changes are managed via a system called **Master Data Governance (MDG)**, which is an add-on to SAP. This program is where you enter the change request, it then applies a review and approval workflow to the request before it gets implemented.* Refer to the Action Plan from the **Weekly Improvement Meeting**. Implement Master Data changes by using **MDG** to raise a request for the change. Changes to Master Data must comply with **E05 Equipment Operating and Maintenance Strategy**. Refer to the following list of QRGs for guidance: • SAP PM QRG – Create & Change IBAU Material • SAP MDG EAM QRG – Bill of Materials MBOM • SAP MDG EAM QRG – Task Lists • SAP MDG EAM QRG – Maintenance Plan Maintenance Item

	• SAP MDG EAM QRG – Measuring Point • SAP MDG EAM QRG – Work Center
When?	Daily as needed

6c Create New Master Data



What?	Create new Master Data.
Why?	**Step 6a** identified where new Master Data needed to be created (for example, an entirely new Maintenance Strategy, or a new Task List, or Work Instruction etc). This step provides an overview of how to create new Master Data.
Who?	• Reliability Engineer • Maintenance Planner • Maintenance Engineer *Note: Responsibility for creating different types of Master Data lie with different people. Please refer to the Roles and Responsibilities page in this document for further guidance*
How?	*Note: Creation of Master Data is managed via a system called Master Data Governance (MDG), which is an add-on to SAP. This program is where you enter the creation request, it then applies a review and approval workflow to the request before it gets implemented.* Refer to the Action Plan from the **Weekly Improvement Meeting**. For actions requiring new Master Data to be created, use **MDG** to raise a request. Creation of new Master Data must comply

	with **E05 Equipment Operating and Maintenance Strategy**. Refer to the following list of QRGs for guidance: • SAP MDG EAM QRG – Functional Location • SAP MDG EAM QRG – Equipment • SAP PM QRG – Create & Change IBAU Material • SAP MDG EAM QRG – Bill of Materials MBOM • SAP MDG EAM QRG – Task Lists • SAP MDG EAM QRG – Maintenance Plan Maintenance Item • SAP MDG EAM QRG – Create Multiple Counter Plan and Item • SAP MDG EAM QRG – Measuring Point • SAP MDG EAM QRG – Work Center
When?	As needed

Weekly Improvement Meeting

Weekly Improvement Meeting	
Purpose	Identify and log opportunities for maintenance and work management continuous improvement.
Timing and Location	Weekly (typically on Wednesday) for approximately 1 hour. Held in a suitable Meeting Room.
Attendees	• Reliability Engineer (meeting lead) • Maintenance Planner • Maintenance Supervisor • Maintenance Superintendent • Maintenance Engineer • Operations Supervisor (optional) • Operations Superintendent (optional)
Preparations	*Note: Refer to **Step 6a** for more details on meeting preparations.* Maintenance Supervisor to collate a list of improvement opportunities from their review of the Work Order history from the previous week (W-1). Refer to **Step 5c Close Work Orders** of sub-process **5. Work Closed Out**). Maintenance Planner to bring relevant maintenance KPI's for the previous week (W-1). Reliability Engineer to bring latest downtime, MTBF, MTTR and other performance and reliability data and analysis for the previous week (W-1) or longer period. Reliability Engineer / Maintenance Engineer to bring any other relevant Root Cause Analysis (RCA) or other defect elimination studies.
Agenda	Review the above data sets, with focus on understanding the reasons for non-compliance. During the analysis, the following must be considered: • Does existing Master Data need to be modified (e.g., Equipment Structure (FLOC), BOM's, inspection frequency, Work Instructions, Task Lists, task durations etc)? • Is a new maintenance strategy needed (e.g. a new service or inspection routine)? • Are any modifications to equipment or assemblies required? • Are there any issues with Work Management workflow, such as the planning and scheduling sub-processes (e.g. blurred responsibilities, non-attendance at meetings), not executing to schedules etc? • Review action plan from the previous meeting. Generate a list of improvement opportunities and add them to the meeting Action Plan. Review all other uncompleted actions.
Meeting Outcomes	List of actions to improve maintenance strategies, practices, work management and Master Data.

Measures of Success

The following are measures of success for sub-process **6. Work Management Improvement**, as described in the Glencore Copper Asset Management Manual **Maintenance Work Management** (Element 08).

Approved Master Data Requests	
KPI Number	KPIGPM012
Definition	Total number of Master Data requests approved as a percentage of all Master Data requests in a period.
Performance Indicator	>90%
Calculation	$\frac{\text{Master Data requests approved}}{\text{Total number of Master Data requests}} \times 100$
Rationale	This is important because rejected Master Data requests increase rework (non-value add work) and also reduces the timeliness of the implementation of new and updated work strategies.

Effective Resource Utilisation (%)	
KPI Number	KPIGPM013
Definition	This KPI represents the total actual hours time confirmed on all work orders as a percentage of the labour capacity available in the same scheduling period.
Performance Indicator	>90%
Calculation	$\frac{\text{Hours confirmed on all work orders}}{\text{Total labour capacity available hours}} \times 100$
Rationale	This is important to monitor the percentage of available hours being captured on work orders. The transparency of this information allows for focus area's to be identified and actioned appropriately to improve work management. *Note: Total labour capacity includes those hours defined as Productive Time, i.e. travel to and from job, job prep, isolation, tool time, work admin etc. It does include non-Productive Time being pre-start/deployment meetings, meal breaks, training etc. An example is illustrated by having 10.5hrs available capacity for a rostered 12hr shift.*



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